DevopsPilot AI

Contents

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Executive Summary

- **Purpose -** DevOps Pilot AI is a centralized project and task management solution built using a Model-Driven App and Microsoft Copilot Studio to simplify and automate the complete workflow of sprint and task management. The application was designed to solve common business challenges such as scattered task tracking, delayed approvals, lack of real-time visibility, and inefficient manual communication between managers and employees.

  The system provides a structured and standardized process where managers can create projects, assign tasks, monitor progress, and approve work efficiently, while employees can easily track and update their assigned tasks through a guided workflow. By integrating Power Automate, the application sends instant notifications and approval requests automatically, reducing manual effort and improving team productivity. The platform also includes role-based access control (RBAC) to ensure secure data handling, allowing managers and employees to access only the information relevant to their roles. Interactive dashboards and visual reports provide clear insights into project status, pending approvals, and overall team performance. Additionally, the integration of AI through Copilot Studio enhances user experience by enabling intelligent assistance and automated responses for task-related queries. The solution is accessible across Web, Tablet, and Mobile devices, making project collaboration faster, more transparent, and highly efficient for modern teams.

- **Background -** The DevOps Pilot AI application was developed to address the growing challenges organizations face in managing projects, approvals, and task tracking efficiently. In many teams, project updates, approvals, and communication were handled manually through Excel sheets, chats, emails, and disconnected tools. This approach often resulted in delayed approvals, lack of transparency, scattered information, and difficulty in monitoring overall project progress. Team members frequently struggled to follow a standardized workflow, while managers lacked real-time visibility into task completion and employee performance.

  To overcome these limitations, DevOps Pilot AI was designed as a centralized and intelligent project management solution using Microsoft Model-Driven Apps, Power Automate, Dataverse, and Copilot Studio .The application enables managers to efficiently assign tasks, review updates, and approve requests through automated processes, while employees can easily access and manage their assigned work in a guided and user-friendly environment.

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Goals and Objectives

- ### Business Goals
- Centralize project, task, and approval management into a single platform for better organization and visibility.
- Automate task approvals, notifications, and workflows to reduce manual effort and delays.
- Improve collaboration and transparency between managers and employees through real-time task tracking.
- Enhance productivity and accountability using dashboards, role-based access control (RBAC), and structured workflows.
- Integrate AI-powered assistance through Copilot Studio to simplify task handling and improve user experience.

- ### KPIs & Success Metrics:

- **Task Completion Rate:** Percentage of tasks completed within the assigned deadline.
- **Approval Turnaround Time:** Average time taken for managers to approve or reject tasks.
- **Project Progress Visibility:** Real-time tracking of completed, pending, and delayed tasks through dashboards.
- **Workflow Efficiency:** Reduction in manual emails, follow-ups, and approval delays after automation.
- **User Adoption & Productivity:** Number of active users and improvement in employee task management efficiency.

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Scope

- ### In Scope:

Key Functional Features -

- Project & Task Management : Managers can create projects, assign tasks, define deadlines, and monitor task progress through a centralized system.
- Approval Workflow Automation
  Automated approval requests and notifications are sent using Power Automate, allowing managers to approve or reject tasks efficiently.
- Real-Time Dashboard & Monitoring
  Interactive dashboards provide real-time visibility into project status, pending approvals, completed tasks, and employee performance.
- Role-Based Access Control (RBAC)
  Managers and employees have secure access based on their assigned roles, ensuring controlled visibility and data security.
- Business Process Flow (BPF) Integration
  Step-by-step guided workflows help users follow standardized processes for task updates and approvals.
- Cross-Platform Accessibility
  The application is accessible on Web, Tablet, and Mobile devices for seamless project management from anywhere.
- AI Integration with Copilot Studio
  AI-powered assistance is integrated to support intelligent responses, task-related queries, and future automation capabilities.

Future Scope:

- Advanced AI-based task recommendations and predictive analytics.
- Integration with additional Microsoft services and third-party tools.
- Enhanced reporting and performance analytics dashboards.
- Voice-enabled AI assistance and smart automation features.

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- **Target Users & Personas:**

- **Employees / Team Members**
  - Primary users responsible for updating work progress and completing assigned tasks.
  - Need a simple and structured workflow to manage daily activities efficiently.
  - Benefit from real-time task visibility, notifications, and streamlined communication.
- **Team Leads / Domain Leads**
  - Manage and monitor tasks assigned to team members within their respective domains.
  - Review progress updates, track workload distribution, and ensure timely completion of activities.
  - Require visibility into task dependencies, approvals, and employee performance.
- **Project Managers**
  - Oversee overall project execution, sprint planning, and task allocation.
  - Create projects, assign responsibilities, define timelines, and monitor project health through dashboards.
  - Need centralized visibility into project progress, approvals, and team productivity.
- **Managers / Approvers**
  - Responsible for reviewing and approving submitted work updates and requests.
  - Monitor employee activities and ensure compliance with organizational workflows and standards.
  - Benefit from automated approval notifications and real-time reporting.
- **Admins / Super Admins**
  - Manage system configurations, user roles, permissions, and workflow settings.

- Maintain platform security, data access control, and overall application governance.
- Require complete visibility into system operations, dashboards, and reports.

- **Executives / Business Stakeholders**
  - Use dashboards and reports to monitor project performance and operational efficiency.
  - Analyze project progress, team productivity, and approval status for strategic decision-making.
  - Require high-level insights and real-time business visibility across projects and teams.

User Stories / Use Cases

ID	Title	User Story
US-1	Project Creation	As a Manager, I want to create projects and define project details so that tasks can be organized and managed efficiently.
US-2	Task Assignment	As a Manager, I want to assign tasks to employees with priorities and deadlines so that work can be distributed properly.
US-3	Task Management	As an Employee, I want to view and manage my assigned tasks so that I can complete my work effectively.
US-4	Work Update Submission	As an Employee, I want to submit daily work updates and progress logs so that managers can track my progress in real time.
US-5	Approval Workflow	As a Manager, I want to approve or reject submitted work updates so that task quality and progress can be monitored properly.
US-6	Real-Time Dashboard	As a Manager, I want to view dashboards and reports so that I can monitor project status, pending tasks, and employee performance.
US-7	Role-Based Access Control	As a User, I want access only to the data and features assigned to my role so that

		information remains secure and organized.
US-8	Automated Notifications	As a User, I want to receive automated email notifications for approvals and updates so that communication becomes faster and more efficient.
US-9	Work Log Tracking	As a Manager, I want to monitor employee work logs and hours worked so that productivity and task completion can be analyzed.
US-10	AI Assistance Integration	As a User, I want AI-powered assistance through Copilot Studio so that I can get quick responses and task-related support.

Functional Requirements

ID	Category	Requirement
FR-1	Project Management	The system shall allow Managers to create and manage projects with project names, descriptions, and timelines.
FR-2	Project Management	The system shall maintain centralized project records using Dataverse tables.
FR-3	Task Management	The system shall allow Managers to assign tasks to employees with priorities, milestones, and due dates.
FR-4	Task Management	The system shall allow Employees to view, update, and manage their assigned tasks.
FR-5	Task Management	The system shall allow users to update task status and submit work progress details.
FR-6	Work Log Management	The system shall maintain daily work logs including hours worked, task details, and employee information.
FR-7	Approval Workflow	The system shall send automated approval requests to Managers using Power Automate workflows.

FR-8	Approval Workflow	The system shall allow Managers to approve or reject submitted work updates and requests.
FR-9	Dashboard & Reporting	The system shall provide dashboards and charts for monitoring project progress, pending tasks, and employee performance.
FR-10	Notifications & Alerts	The system shall send automated email notifications and alerts through Outlook integration.
FR-11	User & Role Management	The system shall implement Role-Based Access Control (RBAC) for Managers and Employees.
FR-12	User & Role Management	The system shall restrict users to access only authorized data and application features based on their roles.
FR-13	AI Integration	The system shall integrate Copilot Studio AI capabilities for intelligent task assistance and query handling.
FR-14	Cross-Platform Accessibility	The system shall support access through Web, Tablet, and Mobile devices.
FR-15	Data Management	The system shall securely store project, task, user, and work log information in Microsoft Dataverse.
FR-16	Real-Time Monitoring	The system shall provide real-time visibility into task progress, approvals, and project activities.

Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	The system shall provide real-time task updates and dashboard visibility with minimal response delay.
NFR-2	Performance	The system shall efficiently handle multiple users, projects, and work log records simultaneously.
NFR-3	Scalability	The application shall support future expansion for additional users, projects, workflows, and AI features.
NFR-4	Availability	The system shall be accessible through Web, Tablet, and Mobile platforms with high availability.
NFR-5	Reliability	The system shall ensure secure and reliable storage of project, task, and work log data in Microsoft Dataverse.
NFR-6	Security	The application shall implement Role-Based Access Control (RBAC) to restrict unauthorized access to data and features.
NFR-7	Security	The system shall securely manage approvals, notifications, and user information within the Microsoft ecosystem.
NFR-8	Usability	The application shall provide a user-friendly and intuitive interface for Managers and Employees.
NFR-9	Maintainability	The system shall support easy modification and maintenance of workflows, forms, dashboards, and business rules.
NFR-10	Integration	The system shall integrate seamlessly with Power Automate, Outlook, Dataverse, and Copilot Studio services.
NFR-11	Compatibility	The application shall support major modern browsers and Microsoft platform services.

NFR-12	AI Capability	The system shall support AI-powered assistance and future intelligent automation through Copilot Studio integration.

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System Architecture

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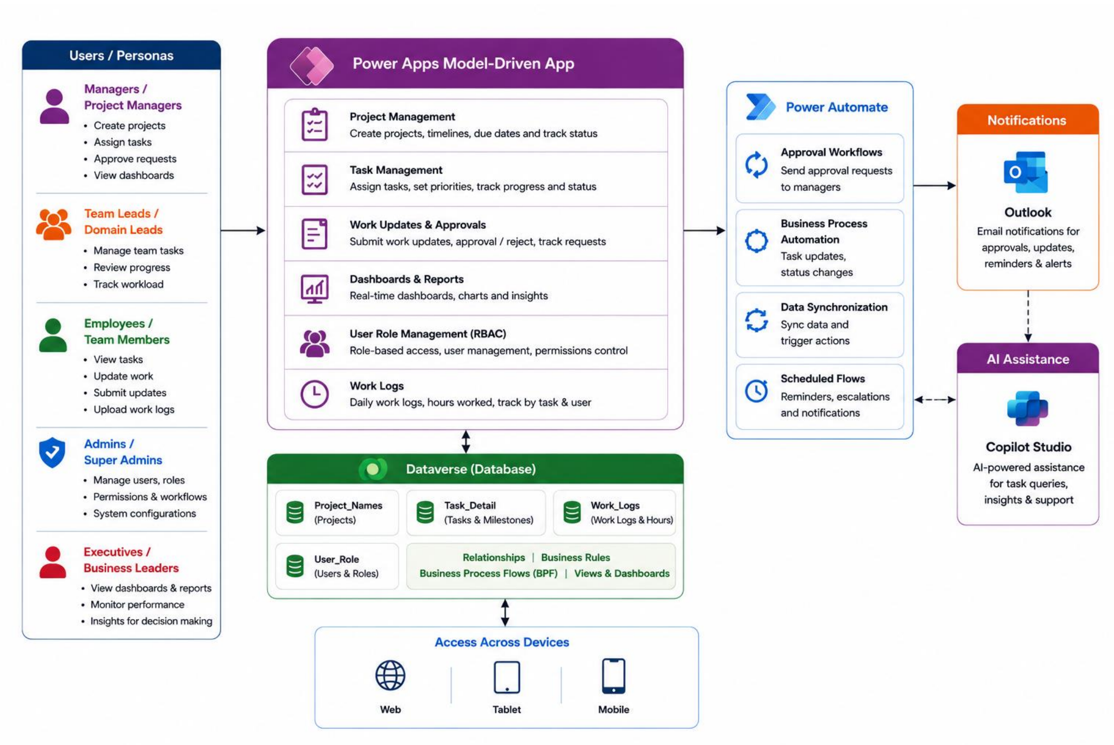
- Overview Diagram



- **Technology Stack Used**
  1) Microsoft Power Apps
  2) Dataverse
  3) Power Automate
  4) Copilot

- **Key Points**

1) User Authentication & Role Management
2) Project Creation & Management
3) Task Assignment & Tracking
4) Work Update Submission
5) Approval & Rejection Workflow
6) Real-Time Dashboards & Reporting
7) Employee Work Log Management
8) Role-Based Access Control (RBAC)
9) Automated Email Notifications
10) Business Process Flow (BPF) Integration
11) Power Automate Workflow Automation

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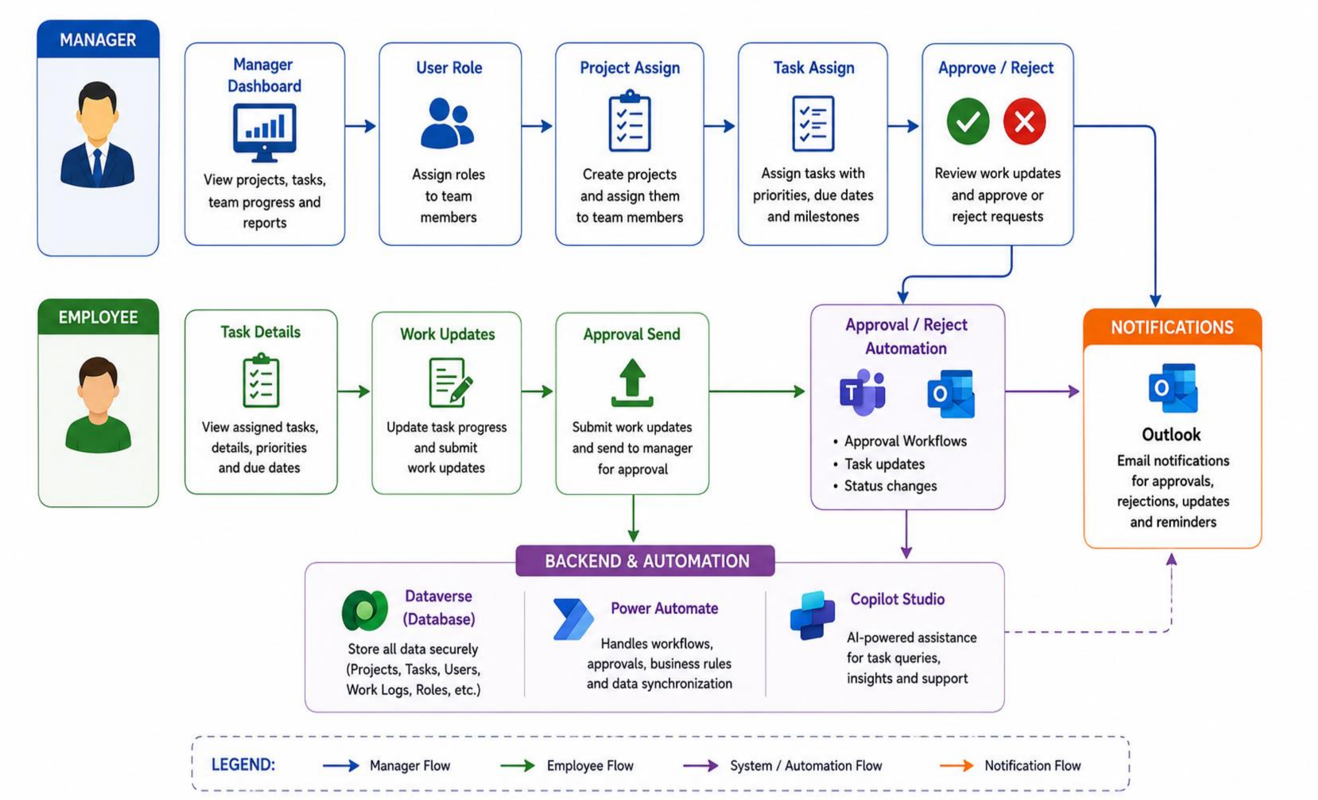
Data Model & Storage

- Data Flow Diagram



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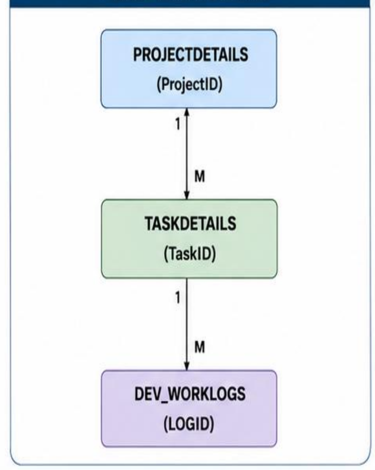
- **Database Schema:**

1. PROJECTDETAILS TABLE				
Field Name	**Data Type**	**Description**	**Key**	**Mandatory**
ProjectID	Number (Auto Number)	Unique Project ID	Primary Key	Yes
ProjectName	Single Line of Text	Name of Project	-	Yes
Description	Multiple Lines of Text	Project Description	-	No
Domain	Choice	Project Domain	-	Yes
StartDate	Date and Time	Project Start Date	-	Yes
EndDate	Date and Time	Project End Date	-	No
ProjectManager	Person or Group	Assigned Project Manager	-	Yes
Status	Choice	Project Status (Active/Completed/Hold)	-	Yes

2. TASKDETAILS TABLE				
Field Name	**Data Type**	**Description**	**Key**	**Mandatory**
TaskID	Number (Auto Number)	Unique Task ID	Primary Key	Yes
TaskName	Single Line of Text	Name of Task	-	Yes
ProjectName	Lookup (ProjectDetails)	Linked Project	Foreign Key	Yes
AssignedTo	Person or Group	Assigned Employee	-	Yes
TaskPriority	Choice	Priority Level (High/Medium/Low)	-	Yes
Start_Date	Date and Time	Task Start Date	-	Yes
End_Date	Date and Time	Task End Date	-	Yes
TaskStatus	Choice	Task Status (In-Progress/Completed)	-	Yes
Description	Multiple Lines of Text	Task Description	-	No

3. DEV_WORKLOGS TABLE				
Field Name	**Data Type**	**Description**	**Key**	**Mandatory**
LOGID	Number (Auto Number)	Unique Worklog ID	Primary Key	Yes
WorkingTask	Lookup (TaskDetails)	Related Task	Foreign Key	Yes
Developer	Person or Group	Employee Name	-	Yes
Worked_Hours	Number	Total Working Hours	-	Yes
LOG_DATE	Date and Time	Work Log Date	-	Yes
LOG_STATUS	Choice	Work Status (Completed/In Progress)	-	Yes
Remarks	Multiple Lines of Text	Daily Work Remarks	-	No

ENTITY RELATIONSHIPS



RELATIONSHIP EXPLANATION		
From Table	**To Table**	**Relationship**
ProjectDetails	TaskDetails	One-to-Many (1:M)
TaskDetails	DEV_WorkLogs	One-to-Many (1:M)

SCHEMA SUMMARY

- **ProjectDetails** stores overall project information.
- **TaskDetails** stores tasks under each project and assigned to employees.
- **Dev_WorkLogs** stores daily work logs and hours spent by employees on tasks.
- One Project can have Multiple Tasks.
- One Task can have Multiple Work Logs.